



ERD Description

1. Users

- user_id (PK) – Unique identifier
- first_name – User's first name
- last_name – User's last name
- email – User's email address
- password – Hashed password
- created_at – Account creation date
- role – Enum (customer, business, administrator)
- shopping_cart (JSON or serialized collection) – Array of product objects with:
 - product_id – ID of the product in the cart
 - unit_price – Price at the time added to the cart
 - quantity – Quantity added
- transactions (JSON or serialized collection) – Array of transaction IDs linked to the user
- addresses (JSON or serialized collection) – Array of address objects with:
 - address_id – Unique identifier for the address
 - street – Street name and number
 - city – City name
 - state – State or province
 - postal_code – ZIP or postal code
 - country – Country name
 - isDefault – Boolean (true if the current address is the default)
- payment_methods (JSON or serialized collection) – Array of payment method objects with:
 - payment_id – Unique identifier for the payment method
 - card_number – Masked card number (e.g., **** * 1234)



- expiry_date – Card expiration date
 - payment_type – Enum (credit_card, debit_card, paypal, etc.)
 - isDefault – Boolean (true if it is the default payment method)
- privileges (JSON object of objects) – Object containing multiple privilege objects as properties, with, for example:
 - shopping:
 - isGranted – Boolean (true if the privilege is granted, false if revoked)
 - description – Description of the privilege
 - granted_by – user_id of the admin or business that granted the privilege
 - granted_at – Date and time the privilege was granted

 Possible Privileges:

- shopping – Allows the user to make purchases
- productManagement – Allows the user to manage products
- userManagement – Allows the user to manage other users
- reporting – Grants access to reports and analytics
- discountManagement – Allows the user to create and modify discount codes
- orderFulfillment – Grants the ability to process and fulfill orders
- customerSupport – Allows access to customer support tools
- marketing – Grants access to marketing campaigns



✓ Privileges JSON Example:

```
{
  "shopping": {
    "isGranted": true,
    "description": "Allows the user to make purchases",
    "granted_by": "admin_123",
    "granted_at": "2025-03-20T10:45:00Z"
  },
  "productManagement": {
    "isGranted": false,
    "description": "Allows the user to manage products",
    "granted_by": "admin_456",
    "granted_at": "2025-03-20T11:30:00Z"
  },
  "userManagement": {
    "isGranted": true,
    "description": "Allows the user to manage other users",
    "granted_by": "admin_789",
    "granted_at": "2025-03-21T09:15:00Z"
  },
  "reporting": {
    "isGranted": true,
    "description": "Grants access to reports and analytics",
    "granted_by": "admin_456",
    "granted_at": "2025-03-22T14:00:00Z"
  }
}
```



2. Products

- product_id (PK) – Unique identifier
- name – Product name
- description (JSON object) – Object with:
 - short – Short description for preview displays
 - long – Detailed description for the product page
 - specs – Specifications (e.g., dimensions, weight, color)
 - features – Array of key product features
- price – Current product price
- stock_quantity – Available stock
- category_id (FK) – Links to Categories
- business_id (FK) – Links to the business that sells the product

✓ Example of the description field in JSON format:

```
{
  "short": "High-quality wireless headphones",
  "long": "These wireless headphones offer exceptional sound quality with noise cancellation, long battery life, and a comfortable fit.",
  "specs": {
    "battery_life": "20 hours",
    "weight": "250g",
    "color": "black"
  },
  "features": [
    "Bluetooth 5.0",
    "Noise cancellation",
    "Fast charging"
  ]
}
```



3. Categories

- category_id (PK) – Unique identifier
 - category_name – Name of the category
 - description – Detailed description of the category
-

4. Transactions

- transaction_id (PK) – Unique identifier
 - user_id (FK) – Links to Users
 - total_amount – Total amount paid
 - payment_method_id – References the payment method used
 - transaction_date – Date and time of purchase
 - shipping_address_id – References the shipping address
 - billing_address_id – References the billing address (defaults to the shipping address but can be different)
 - products (JSON or serialized collection) – Array of product objects with:
 - product_id – ID of the product purchased
 - unit_price – Price at the time of purchase
 - quantity – Quantity purchased
 - tracking_status (Stack collection of tracking objects) – Array of tracking objects representing the delivery status stack, for example:
 - tracking_id – Unique identifier for the tracking event
 - status – Enum (purchased, shipped, in_transit, out_for_delivery, delivered, cancelled – by customer or admin, return, returned – by customer.)
 - status_date – Date and time of the status update
 - latitude – Google Maps latitude coordinate
 - longitude – Google Maps longitude coordinate
-